CDA 106
Numerical Methods for Data Science

Dr Prashant Kumar Srivastava

IIT Patna

pksri@iitp.ac.in

LECTURE-14 | March20, 2026

$$PA = LU$$

* we used <u>elementary matrices</u> to find PLU decomposition

Consider

$$A = \begin{bmatrix} 4 & 4 & 4 & 4 \\ 1 & 1 & 2 & 2 \\ 4 & 5 & 5 & 6 \\ 2 & 4 & 4 & 4 \end{bmatrix}$$

Aim: To find $PA = LU$

$$P_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

$$A \xrightarrow[\substack{R_3 - R_1 \\ R_4 - \frac{1}{2}R_1}]{R_2 - \frac{1}{4}R_1} \begin{bmatrix} 4 & 4 & 4 & 4 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 2 & 2 & 2 \end{bmatrix} \xrightarrow{R_4 \leftrightarrow R_2}$$

$$\xrightarrow{R_4 \leftrightarrow R_2} \begin{bmatrix} 4 & 4 & 4 & 4 \\ 0 & 2 & 2 & 2 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \end{bmatrix} \xrightarrow{R_3 - \frac{1}{2}R_2} \begin{bmatrix} 4 & 4 & 4 & 4 \\ 0 & 2 & 2 & 2 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\xrightarrow[P_2]{R_4 \leftrightarrow R_3} \begin{bmatrix} 4 & 4 & 4 & 4 \\ 0 & 2 & 2 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \leftarrow U$$

$$P_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1/2 & 1 & 0 & 0 \\ +1 & +\frac{1}{2} & 1 & 0 \\ 1/4 & 0 & 0 & 1 \end{bmatrix}$$

(Intermediate)

$$\downarrow$$

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1/2 & 1 & 0 & 0 \\ 1/4 & 0 & 1 & 0 \\ 1 & 1/2 & 0 & 1 \end{bmatrix}$$

$$P = P_2 P_1$$

$$\boxed{PA = LU}$$

Verify this: $LU = \begin{bmatrix} 4 & 4 & 4 & 4 \\ 2 & 4 & 4 & 4 \\ 1 & 1 & 2 & 2 \\ 4 & 5 & 5 & 6 \end{bmatrix} = PA = P_2 P_1 A$

For a matrix A to have L U decomposition

- All the leading principle minors are non-zero (Necessary & Sufficient)

- Matrix A is positive definite.

"very imp. class of matrices."

Positive Definite Matrix

A real-symmetric $n \times n$ matrix is said to be positive definite if for every $x = (x_1, x_2, x_3, \ldots, x_n)^T$, $x \in \mathbb{R}^n \setminus \{0\}$

$$x^T A x > 0$$

Imp. prop ① Positive definite matrices have all eigen values as positive.

② If A is positive definite then all its principle minors will be positive.

$Ax = \lambda x \qquad x \neq 0$

$x^T A x > 0 \Rightarrow x^T \lambda x > 0 \Rightarrow \lambda (x_1\ x_2 \cdots x_n) \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} > 0 \Rightarrow \lambda \overbrace{(x_1^2 + x_2^2 + \cdots + x_n^2)} > 0 \Rightarrow \lambda > 0$

Cholesky decomposition (factorization) [Square root factorization]

$A_{n \times n}$ — Symmetric
— positive definite

Square matrix $A_{n \times n}$

Symmetric

$A^T = A$

or $\boxed{a_{ji} = a_{ij}}$

$\forall i \& j$

$i \neq j$

$$A = LL^T$$

or

$$A = UU^T$$

Example

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 8 & 22 \\ 3 & 22 & 82 \end{bmatrix}$$

Aim is to find L s.t. $A = LL^T$

(Note that A is positive definite)

Consider $L = [l_{ij}]_{3\times 3}$ $\quad l_{ij} = 0 \quad i < j$

writing

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 8 & 22 \\ 3 & 22 & 82 \end{bmatrix} = A = LL^T = \underbrace{\begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} l_{11} & l_{21} & l_{31} \\ 0 & l_{22} & l_{32} \\ 0 & 0 & l_{33} \end{bmatrix}}_{L^T}$$

$l_{11}^2 = 1$, $\quad l_{11} l_{21} = 2 \quad l_{11} l_{31} = 3$

$l_{21} l_{11} = 2 \quad l_{21}^2 + l_{22}^2 = 8 \quad l_{21} l_{31} + l_{22} l_{32} = 22$

$l_{31} l_{11} = 3 \quad l_{31} l_{21} + l_{32} l_{22} = 22 \quad l_{31}^2 + l_{32}^2 + l_{33}^2 = 82$

6 eqns in 6 unknowns

$$l_{11}^2 = 1 \Rightarrow \boxed{l_{11} = 1} \quad (\text{we choose})$$

$$\boxed{l_{21} = 2} \qquad \boxed{l_{31} = 3}$$

$$l_{21}^2 + l_{22}^2 = 8 \Rightarrow l_{22}^2 = 8 - 4 = 4 \Rightarrow \boxed{l_{22} = 2} \quad (\text{we choose})$$

$$l_{21} l_{31} + l_{22} l_{32} = 22 \Rightarrow \underline{\underline{l_{32}}} = \frac{22 - 2 \times 3}{2} = \underline{\underline{8}}$$

$$l_{31}^2 + l_{32}^2 + l_{33}^2 = 82$$

$$l_{33}^2 = 82 - 9 - 64 = 9 \Rightarrow \boxed{l_{33} = 3} \quad (\text{we choose})$$

<u>Therefore</u>

$$A = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 3 & 8 & 3 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 8 \\ 0 & 0 & 3 \end{bmatrix}}_{L^T} = \underline{\underline{LL^T}} \quad \left[\text{Cholesky decomposition}\right]$$

Consider the system of eqn

$$\left.\begin{aligned} x_1 + 2x_2 + 3x_3 &= 5 \\ 2x_1 + 8x_2 + 22x_3 &= 6 \\ 3x_1 + 22x_2 + 82x_3 &= -10 \end{aligned}\right]$$

$x_1 = 2$
$x_2 = 3$
$x_3 = -1$

$$\underbrace{\begin{bmatrix} 1 & 2 & 3 \\ 2 & 8 & 22 \\ 3 & 22 & 82 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{X} = \underbrace{\begin{bmatrix} 5 \\ 6 \\ -10 \end{bmatrix}}_{b}$$

$\underline{\underline{AX = b}}$ we use Cholesky decomposition &

Write $A = LL^T$

Where $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 3 & 8 & 3 \end{bmatrix}$ [we got it from example on slide 8]

Therefore, $AX = b$ can be written as

$L(L^T X) = b$ $\qquad Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$

$\Rightarrow LY = b$ (Forward Substitution)

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 3 & 8 & 3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \\ -10 \end{bmatrix}$$

$\boxed{y_1 = 5}$

$2y_1 + 2y_2 = 6 \Rightarrow \boxed{y_2 = -2}$

$3y_1 + 8y_2 + 3y_3 = -10 \Rightarrow \boxed{y_3 = -3}$

$$\overset{L^T}{\begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 8 \\ 0 & 0 & 3 \end{bmatrix}} \overset{X}{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ -3 \end{bmatrix}$$

$\Rightarrow$ Backward Substitution $\Big\}$ $x_3 = -1$, $x_2 = 3$, $x_1 = 2$!!

So far we have studied

- LU decomposition
- Cholesky decomposition
- PLU decomposition
- QR decomposition

  (This we shall study now)

Eigen Value problem

$A \longrightarrow$ square matrix $n \times n$

$\underline{\underline{x \neq 0}}$ $\quad x = (x_1, x_2, \ldots, x_n)^T$ eigen vector

if for some value 'λ'

$\underline{\underline{Ax = \lambda x}}$, and λ – eigen value.

Que How to find eigen values & eigen vectors !!

$$\underset{n\times n}{A}X = \lambda X \qquad \text{(eigen value problem)}$$

$$\Rightarrow A_{n\times n} X_{n\times 1} = \lambda I_{n\times n} X_{n\times 1} \qquad I_n = I_{n\times n}$$

$$\Rightarrow \underbrace{\underbrace{(A-\lambda I_n)}_{B} X_{n\times 1} = 0}_{BX=0 \text{ a system of homogeneous eq}^n}$$

zero vector

$0 = (0, 0, \ldots, 0)$

X ← unknown

Therefore, for non-zero X, we must have $|B| = 0$ or

$$|A - \lambda I_n| = 0 \longleftarrow \text{characteristic eq}^n$$

λ is a root of the characteristic eq^n

& $(A-\lambda I)X = 0$, $X \neq 0$ becomes corresponding eigen vector.

Example

$$A_{2\times 2} = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$$

Aim: To find eigen values & eigen vectors

$$A x = \lambda x$$

? ?

Ch. eqn

$$|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 1-\lambda & 2 \\ 3 & 2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda)(2-\lambda) - 6 = 0$$

$$\Rightarrow \lambda^2 - 3\lambda + 2 - 6 = 0$$

$$\Rightarrow \lambda^2 - 3\lambda - 4 = 0 \longrightarrow \text{quadratic eqn}$$

$$\Rightarrow \boxed{\lambda = 4}, \boxed{\lambda = -1}$$

eig values.

For eigenvector corresponding to $\lambda = 4$

$$(A - 4I)X = 0$$

$$\Rightarrow \underbrace{\begin{bmatrix} -3 & 2 \\ 3 & -2 \end{bmatrix}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\Rightarrow -3x_1 + 2x_2 = 0 \longrightarrow \text{one eq}^n \text{ in 2 variables}$$

$$x_1 = 2 \Rightarrow x_2 = 3$$

$\lambda_1 = 4$ $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$

Similarly for $\lambda_2 = -1$, $\underbrace{\begin{bmatrix} 1 \\ -1 \end{bmatrix}}_{\text{Check (?)}}$

If the size of matrix A is very large

$A_{n \times n} \longrightarrow$ n is very large

Your Ch. eqn is a polynomial of degree 'n'

and, in general, we cannot expect to solve explicitly

the polynomial eqns of ~~der~~ higher degrees (≥ 5)

In such cases, we use some of the iterative techniques ~~to~~ find eigen values.